

The Expectations of Project Managers from Artificial Intelligence: A Delphi Study

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Abstract

Artificial intelligence (AI) technologies are rapidly developing these days and are expected to impact the field of project management on multiple levels; however, there remains a high level of uncertainty regarding the effect that AI might have on project management practices. This article aims to address this topic based on a Delphi study with a panel of 52 project management experts who reflected on future potential AI applications for the project management Knowledge Areas. The article provides a visionary perspective that can be further translated into practical solutions in the near and far future to improve project management practices.

Keywords

artificial intelligence, project management, Delphi study

Introduction

The latest developments in artificial intelligence (AI) technologies are expected to radically impact the economy and dramatically change the business landscape and overall workplace environment and job responsibilities. Recent studies by consulting companies demonstrate those forecasts. For example, a study by PricewaterhouseCoopers estimated that these technologies will contribute up to 15% of the global GDP (gross domestic product) by 2030 (Hawksworth et al., 2018). McKinsey studies estimated that about one-third of the constituent activities could be automated by 2030, which will transform the workplace and have a dramatic impact on work displacement (Manyika et al., 2017), and that there is nearly a 25% year-over-year increase in the use of AI (McKinsey Analytics, 2019). There is much uncertainty, however, with regard to the ways in which AI will impact organizations, managers, and employees (Daugherty & Wilson, 2018; Iansiti & Lakhani, 2020; Raisch & Krakowski, 2020; Ransbotham et al., 2017); this uncertainty also exists in the discipline of project management (Auth et al., 2019; Lahmann, 2019; Munir, 2019; Parsi, 2019; Project Management Institute [PMI], 2019; Wang, 2019). Throughout the project lifetime, a huge amount of data is collected and analyzed by new technologies and information systems, mainly for purposes of monitoring and controlling and improved decision-making. However, with the increase of project complexity on the one hand (Afzal et al., 2019; Bakhshi et al., 2016; Jaafari, 2003) and the capabilities of new technologies to support advanced performance on the other hand (Lee & Shin, 2020; Makridakis, 2017; Paschen et al., 2020), there is a need to see how these trends can

be joined for the benefit of more effective project practices, which will, in turn, lead to more successful projects. In many of the previous studies on AI in project management, attention was given to processes based on numerical data, such as effort estimation, budgeting, and cost management (e.g., Afzal et al., 2019; Cheng et al., 2015; Kamoona & Budayan, 2019; Nassif et al., 2019); schedule management (e.g., Costantino et al., 2015; Faghihi et al., 2015; Jin & Zhang, 2011; Khodakarami & Abdi, 2014; Y.-R. Wang et al., 2012); and risk management (Costantino et al., 2015; Hartono, 2018; Jin & Zhang, 2011; Qazi et al., 2016). Yet, the overall management of projects calls for a comprehensive perspective that encompasses a wide range of practices and processes in different Knowledge Areas. Therefore, for practitioners and scholars currently in the field of project management, the debate is not only about the technological feasibility of AI beneficial solutions (Kolbjørnsrud et al., 2016) and related ethical issues (Haenlein & Kaplan, 2019), but also in the ways AI will be able to support the variety of managerial processes. The motivation for the current study derives from the positions of project managers in complex projects and the need for support in performing numerous tasks that are under their responsibility (Bakhshi et al., 2016). In this article, we highlight and present

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the domains in which project managers think new technologies can improve their work.

With the increasing trend of projectification, where projects become a form of business organization (Maylor & Turkulainen, 2019; Schoper & Ingason, 2019), there is a need to explore the opportunities that AI may offer to project managers and project teams. Since this is a new arena for both researchers and practitioners, this study is based on an explorative approach that aims to answer the question of what the expectations are of current project managers from future AI applications, and therefore to provide outlines for the preferred developments of AI in project management. The study presents potential AI systems and applications as perceived by project managers, thus representing the future users' needs and expectations. Although it is beyond the scope of this article to provide an extensive review of AI, it starts with a brief review of the history and characteristics of AI technologies. The following section describes the three-round Delphi study, employed to collect consensual replies from a wide population of experienced project managers. The discussion includes interpretation of the results and deals with additional issues that emerged during the study. The article concludes with a summary concerning the theoretical and practical implications, the research limitations, and suggested directions for future research.

AI Technologies

The modern concept of AI was introduced in the 1950s with the seminal work of Alan Turing (Turing, 1950), who suggested that intelligent machines can replicate the reasoning processes done by humans when they use available information to solve problems and make decisions. Since then, the discipline of AI has experienced ups and downs with regard to the level of interest, thanks to several inspiring success stories on the one hand and a mismatch between expectations and realizations on the other (Haenlein & Kaplan, 2019). Recent definitions of AI suggest that they are “systems that display intelligent behaviour by analysing their environment and taking actions—with some degree of autonomy—to achieve specific goals” (European Commission [EC], 2018, p. 1), and that AI is a “system’s ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation” (Kaplan & Haenlein, 2019, p. 17). Although there are various definitions of AI in the literature, the main concept of AI is based on the increasing capabilities of technology to learn how to analyze data and perform tasks that are currently performed by humans.

Multiple AI technologies and techniques have been developed that can be used to perform intelligent decision-making processes and tasks. Conceptually, those methods can be classified based on the level and range of the application. Some technologies are aimed at problem-solving, where a goal-based approach is used to analyze a specific, though possibly complex question or problem that needs a solution (Luger, 2005). In these types of applications, algorithms are used to

identify possible sequences to reach a goal or to find the optimal solution under certain circumstances. Other technologies are based on reasoning and apply deductive or inductive logic techniques that follow patterns or rules to generate conclusions for decision-support systems (Paschen et al., 2020; Poole & Mackworth, 2010). Machine learning technologies facilitate adaptive AI solutions, where patterns are extracted from data with no *a priori* definitions; therefore, the system improves its performance as it learns from the data that are processed (Paschen et al., 2020). Machine learning algorithms that are used for clustering, classification, or prediction, can be categorized as supervised, unsupervised, semisupervised, and reinforcement learning (Lee & Shin, 2020). Deep learning—a subfield of machine learning—is based on artificial neural networks, where hierarchical representation is extracted progressively, thus achieving greater power and flexibility (Goodfellow et al., 2016).

AI systems can be differentiated by level of specialization, where narrow AI (ANI) refers to systems that are able to learn and do only what they were designed for; general AI (AGI) refers to intelligent systems that can learn about any problem and solve it (Deloitte, 2018). In addition, AI systems can be differentiated by level of intelligence, where weak AI refers to systems that are focused on doing one task really well, whereas strong AI refers to systems that are intelligent by nature and self-conscious enough to perform even better than humans in a wide range of tasks (Borana, 2016). The highest level of AI developments—super AI or superintelligence—refers to systems that exceed human intelligence and abilities (Boström & Yudkowsky, 2014; Müller & Boström, 2016), defined by Boström as “intellects that greatly outperform the best current human minds across many very general cognitive domains” (Boström, 2014, p. 52).

AI in Project Management

The literature review on AI in project management reveals that, during the 1990s, the technologies and methods used in projects were mainly expert systems for decision-making and simulations (Arditi & Patel, 1989; Currie & Drabble, 1992; Diekmann, 1992). However, more recent applications refer to neural network models (Chaphalkar et al., 2015; Wang et al., 2012), fuzzy logic, and hybrid models (Afzal et al., 2019; Attar et al., 2013; Cheng & Roy, 2011).

Concerning the professional domains of project management, previous studies mainly addressed the topics of risk management (Taroun & Yang, 2013; Wang & Jin, 2019), scheduling (Currie & Drabble, 1992; Faghihi et al., 2015), effort estimation (Elish et al., 2013; Hosni et al., 2018; Nassif et al., 2019; Predescu et al., 2019), and quality of deliverables (Nemati et al., 2002). Other studies discussed the relationships between risk assessment and the impact on project costs and budgeting (Afzal et al., 2019; Cheng et al., 2015; Cheng & Roy, 2011; Diekmann, 1992; Kamoona & Budayan, 2019; Siqueira, 1999); yet others added to the equation the domain

of schedule (Dawson & Dawson, 1995; Gong, 1997; Leu et al., 2001; Wang et al., 2012). All those project management Knowledge Areas are usually quantitative-based, and therefore more easily translated to AI systems based on computerized analyses. However, later studies addressed less quantitative domains such as procurement processes, mainly related to contract negotiation and disputes (Attar et al., 2013; Chaphalkar et al., 2015; Kog & Yaman, 2016) and project knowledge management and competences (Brahmi & Matta, 2018; Colomo-Palacios et al., 2014; Lin et al., 2015). These developments might indicate that AI has become more advanced and will also be able to support project managers in qualitative-based processes such as communications and stakeholder management.

The current study aims to explore all managerial activities in the project environment that might benefit from AI systems, whether they exist today or will be developed in the future. It is based on the Delphi method, which is a structured process of interactive communication with a panel of experienced project managers to generate ideas and forecast future possibilities.

Delphi Study on AI in Project Management

Qualitative Delphi Method

The Delphi method—named after the Oracle of Delphi—was originally developed by the Rand Corporation in the 1950s as a “method for structuring a group communication process so that the process is effective in allowing a group of individuals, as a whole, to deal with a complex problem” (Linstone & Turoff, 1975, p. 3). This survey technique was adopted for the purpose of the current exploratory study as a useful approach to developing a set of alternative future possibilities and reaching a stable response agreement among a panel of experts with regard to a new complex concept (Grisham, 2009; Habibi et al., 2014; Hsu & Sandford, 2007; Linstone & Turoff, 2011; Pichyangkul et al., 2012). The Delphi technique has been criticized by several scholars throughout the years due to its weaknesses with regard to a basic source of information, limited interaction, vulnerability to manipulations and irresponsible responses, a low level of accuracy, and complications related to research procedures (Landeta, 2006; Woudenberg, 1991). Despite the criticism, when conducted correctly, it is an appropriate method and previously used in the field of business and management to shed light on emerging concepts and practices (Giannarou & Zervas, 2014). Previous studies used the Delphi method in the context of newly developed technologies such as blockchain (White, 2017), big data (Roßmann et al., 2018), and business analytics (Vidgen et al., 2017). As a general rule, a qualitative research approach that is grounded in inputs from experienced and knowledgeable project managers and involves critical interpretive perspective, may yield new insights into complex issues in project management (Cicmil, 2006). A comparison of a traditional survey approach with the Delphi technique indicates that the latter is the appropriate

method to investigate questions of high uncertainty and speculation, where rich data can be obtained from an array of experts who provide their thoughts and opinions (Okoli & Pawlowski, 2004), which is the case in this study. The anonymous procedure, where the respondents do not know one another, coupled with the feedback process that allows them to reassess their initial inputs through iterative sessions, encourages participants to explore new concepts freely and thoroughly (Hsu & Sandford, 2007). The innovativeness of AI—as yet scarcely known and seldom implemented and practiced in most business domains and particularly in project management—rationalizes the use of the Delphi method for the purpose of this research.

Research Design

The current research adopted the guidelines recommended by previous researchers (Grisham, 2009; Habibi et al., 2014; Hasson et al., 2000; Hsu & Sandford, 2007; Landeta, 2006; Okoli & Pawlowski, 2004; Sekayi & Kennedy, 2017) and made some required adjustments to align general procedures with this research setting. It started with identification of the research problem and continued with selection of the expert panel and informing them about the process. Then, three rounds of data collection were then applied, followed by qualitative analyses at the end of the first two rounds and a quantitative analysis at the end of the third round. The process was finalized by interpreting the insights and presenting the results (Figure 1).

Identification of the Research Problem

Following the literature review it was realized that there is a gap in the existing body of knowledge with regard to potential applications of AI in the field of project management. Recent developments in AI seem promising these days in terms of changing the business environment (Cubric, 2020; Gartner, 2020; Hawsworth et al., 2018; McKinsey Analytics, 2019; Ransbotham et al., 2017) and are expected to impact both project management practices and the role of project managers (Auth et al., 2019; Lahmann, 2019; Parsi, 2019; PMI, 2019, 2020; Wang, 2019). Yet, there is only limited academic and practical information on how AI will be applied in those environments.

This study aims to gain insight from project management practitioners' perceptions on how AI will change the world of project management. Therefore, the research question is: *What are the expectations of current project managers from future AI applications?* The general question was broken down to identify *how specific project management processes can benefit from AI*, while taking into consideration each of the ten Knowledge Areas: Project Integration Management, Project Scope Management, Project Schedule Management, Project Cost Management, Project Quality Management, Project Resource Management, Project Communications Management, Project Risk Management, Project Procurement Management, and Project Stakeholder Management (PMI, 2017).

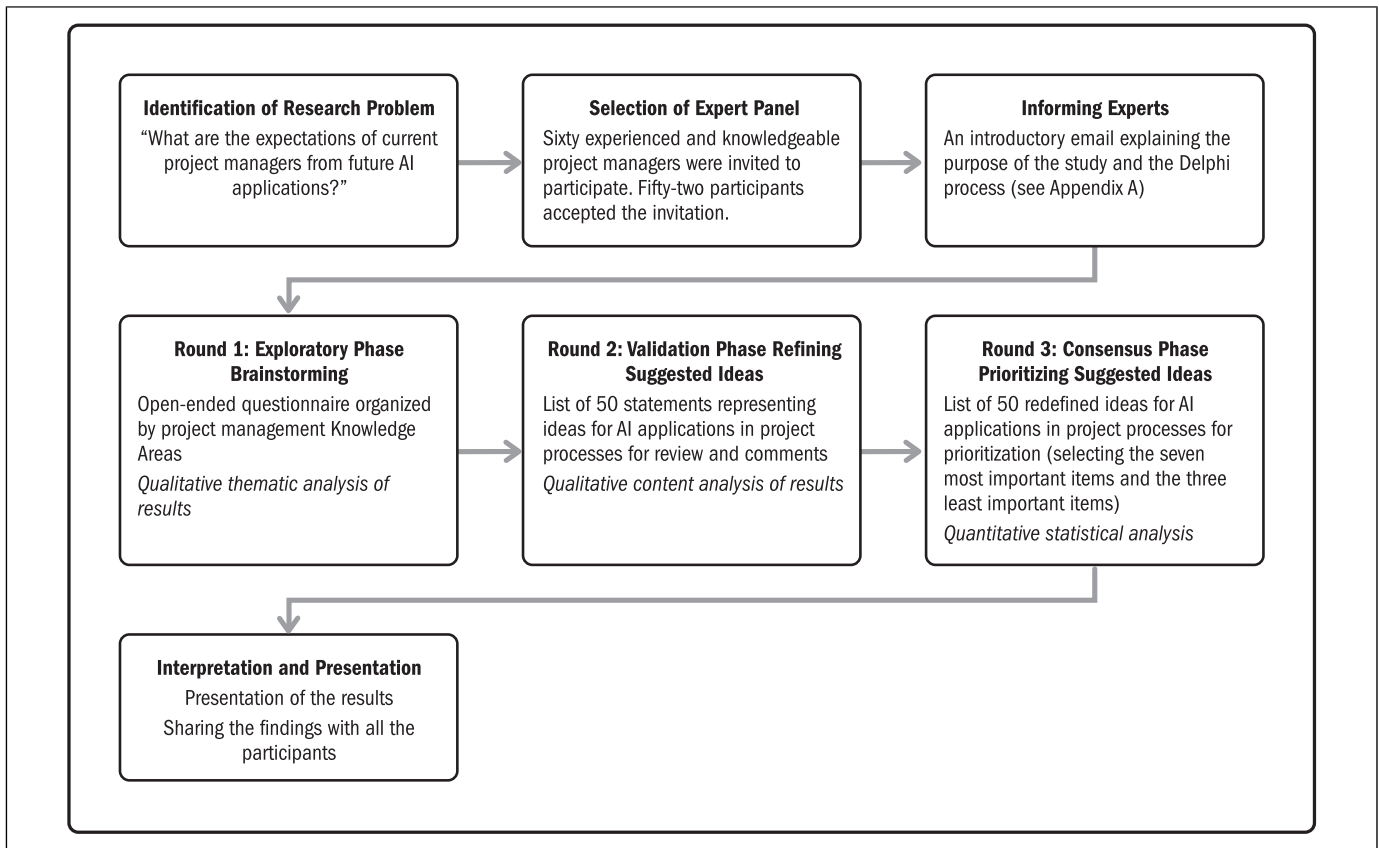


Figure 1. The research design process.

As an exploratory study on potential future AI developments, the current research refrained from defining a closed list of specific options that would have narrowed and limited the responses to quantitative assessments on existing practices and implemented processes. On the contrary, since AI in project management has not been thoroughly studied or investigated in the past, and because it presents a new horizon that is currently undefined, this research builds on the vast experience of project practitioners to acquire new understanding of their comprehensions, perceptions, and visions. However, to obtain a comprehensive, systematic, and structured overview of the potential application of AI in project management processes, the ten different, though related, Knowledge Areas presented in *A Guide to the Project Management Body of Knowledge (PMBOK® Guide)* – Sixth Edition (PMI, 2017), were selected as a reference platform for this study. Many project managers use the *PMBOK® Guide* as a guideline for their daily work, are familiar with the terminology, and find it accessible and easy to understand. As an accepted standard in the project management field, the *PMBOK® Guide* will pave the way for future AI developments that will contribute to improved project processes and eventually to project success. Furthermore, previous studies of the broad project management concepts discussed in the *PMBOK® Guide* will serve as a reference in their studies on skills (Pant & Baroudi, 2008), risk management (Holzmann &

Spiegler, 2011), project maturity (Globerson & Zwikael, 2002; Ibbs & Kwak, 2000), and success (Ehsan et al., 2010; Gomes et al., 2021; Niazi et al., 2016).

The Expert Panel

Selection of the expert panel was conducted using a sequential sampling snowball method based on project management practitioner qualifications with the understanding that, unlike other research methods, the Delphi study does not require a representative sample of a population (Delbecq et al., 1975). The level of qualifications was defined by the number of years of experience as project managers and self-assessed degree of knowledge about project management practices. No qualifications were defined with regard to knowledge of AI, since the study is aimed at understanding the AI future user perspective and expectations, rather than the actual possibilities of using AI today. The participants were unbiased and impartial (Hasson et al., 2000). The current study, which explores ideas and thoughts about global future developments, makes clear that the participants will not be directly affected by the study results.

The number of participants in an expert panel is not defined in the methodological literature and may vary from only a few members to as many as more than 100, as long as they are composed of a heterogeneous group of competent professionals

(Habibi et al., 2014). A group of a few dozen participants is considered a large sample size that is able to produce a high volume of data (Hasson et al., 2000). In this study, 60 experienced and knowledgeable project managers were invited to participate; eight of them were not available at the time of the research, which resulted in 52 participants from the United States, the United Kingdom, and Israel taking an active part in the study during the summer of 2020. Tables 1 and 2 present their distribution by gender and years of experience in the project management field and by industry.

The majority of the panelists have more than 10 years of experience in project management (84.62%), including 25% (13 experts) with more than 20 years of experience, and an additional 25% (13 experts) with more than 30 years of experience in project management. During their extensive experience, many of the project managers have moved from one industry to another, resulting in a wide variety of industries in which the panelists managed projects and gained insights on the needs for improved processes. Therefore, the panel represents a diversified group of knowledgeable and experienced project managers who can provide informed opinions on the needs and desired benefits of applying AI in projects. It is worthwhile emphasizing that it was an intentional decision to include, in the expert panel, project management practitioners with a zero to moderate level of knowledge on AI, but not AI experts, since the focus of this study was to understand the perceptions, expectations, and preferences of future users of AI in project management. A similar approach was previously taken in other Delphi studies, including studies on the impact of digitization on auditing (Tiberius & Hirth, 2019), the role of supply chain management executives (Wehrle et al., 2020), and research on the impact of service robots on leadership and human resource management in the hospitality industry (Xu et al., 2020). The fact that the expert panel did not include AI professionals enabled them to be free from thinking about the limitations of AI technologies today and ignore questions of feasibility.

Data Collection in a Three-Round Process

The participants were first approached via an introductory email in which the context and the purpose of the study were presented, and the guidelines for participation in the process were described. They were also informed of the anonymity of their identity to other participants. A copy of the introduction message is presented in Appendix A.

The three-round process was structured as follows:

Round 1: Brainstorming was aimed at exploring ideas on potential AI applications for project management, based on the participants' knowledge and needs. The participants were asked to share their thoughts and suggest as many ideas as they could think of for AI applications, assuming there were no technological limitations. As AI in project management is a new and unfamiliar domain, the format of open-ended questions allowed the participants complete freedom to generate new and innovative ideas. This step produced a large amount

of data that were analyzed by the researchers using a qualitative thematic method (Brady, 2015). The responses were coded and carefully examined by the researchers to identify common concepts and categorize them according to the ten project management Knowledge Areas (PMI, 2017) and an additional category of "others." The thematic analysis yielded a list of 50 desirable functions of AI applications in project management, each related to a process or general topic in a Knowledge Area.

Round 2: Validation of the first round of responses was aimed at verifying that the responses were correctly interpreted and categorized, to refine the categorization, and to enable the participants to suggest additional ideas that might emerge when reviewing their colleagues' responses (Okoli & Pawlowski, 2004). The panelists received a detailed table, which included consolidated results from round 1 and were asked to review the identified project management functions to be enhanced by AI applications. Next to each function, there was a detailed description of the function, the expected outputs, and the data sources needed to produce them. The participants could comment on each item on the list to refine the idea, suggest additional items, or recommend the deletion of items. The responses were recorded and analyzed by the researchers who used a qualitative content analysis method to ensure that all the additional data were well represented and assigned to the relevant category (Hasson et al., 2000). The detailed list included the functions that could benefit from AI, categorized by process/topic in each Knowledge Area and relevant data sources that could contribute to produce valuable outputs (see Appendix B).

Round 3: Consensus among the panelists was aimed at prioritizing the ideas for future AI applications for improved project management practices by ranking the identified items. There are various methods on how to assess the level of agreement in Delphi studies in order to achieve a consensus among the participants, including Likert scales (Giannarou & Zervas, 2014), rankings (Okoli & Pawlowski, 2004), and endorsements (Sekayi & Kennedy, 2017), which range from 3 to 10 levels. In the current study, because the results of round 2 produced an extensive list of ideas, the panelists were asked to carefully review all the suggested ideas and identify the seven most relevant and important functions ("strongly endorsed") and the three least relevant and important functions ("minimally endorsed").

Results

The most relevant and important project management processes and functions selected by the panelists are presented in Table 3.

The results show that the most endorsed functions are related to the management of schedule, scope, cost, and risk (detailed descriptions are presented in Appendix B). The reason for those high evaluations might be related to the panelists' understanding that these topics are based on quantitative data and

Table 1. Expert Panel by Gender and Years of Experience

Experience	Male		Female		Total	
	Frequency	Percentage	Frequency	Percentage	Frequency	Percentage
<5	-	-	1	5.90	1	1.92
5–10	5	14.30	2	11.80	7	13.46
11–20	12	34.30	6	35.30	18	34.62
21–30	8	22.90	5	29.40	13	25.00
>30	10	28.60	3	17.60	13	25.00
Total	35	100.00	17	100.00	52	100.00

therefore can benefit the most from computerized analyses. However, some of the panelists provided additional explanations for their selections, stating that schedule, scope, cost, and risk are also the most basic processes in project management. Those core processes are perceived as the foundations of project management, and many other highly important functionalities cannot stand on their own without first laying the groundwork for them.

An in-depth investigation of the results reveals that, from the perspectives of practitioners, almost all the processes related to scope management can benefit from AI. Specifically, the practitioners would like to improve the task of extracting the list of deliverables while taking into consideration the contractual dates, technical and other constraints, customers' expectations, expected costs, and quality standards. The data used to produce the list of deliverables should be collected from diverse sources, including communication with the customer, specifications and descriptions of similar products, industrial standards, and internal documents and procedures in the organization. Additional important functions that can utilize AI applications include automated creation of a work breakdown structure (WBS) and the list of tasks to be accomplished during the project's lifetime, along with dynamic prioritization of the tasks, which can be applied when using both waterfall and agile approaches. Project professionals expect AI applications to synthesize technical and financial data along with market trends, availability of resources, and internal historical documentation, in order to produce the project work plan. Furthermore, they

would like AI applications to continuously check and update the priorities of executing the project tasks and to identify, as soon as possible, deviations from the plan and provide analyses of expected implications. Another Knowledge Area considered by project professionals to be significantly enhanced by AI is Project Schedule Management. In this case, the expectations go beyond automated processes to *smart* planning and monitoring. This means that through the integration of multiple sources and formats of data, the project schedule—which is based on feature complexity, available resources, team expertise, as well as internal and external constraints and patterns of work—will be created and further analyzed in order to recommend (and potentially also actually add) buffers, milestones, and stage gates. Once the project is started, the schedule will be updated not only based on progress reports, but also based on visual analyses of the developed product, formal and informal discussions on user stories completion, or analyses of forecasted deviations that may occur due to internal or external factors. The AI application will provide analyzed information on expected implications on multiple levels, including achievement of the project objectives, costs, stakeholder satisfaction, and more. This functionality is also related to the process of risk identification and analysis, where the project experts mentioned the possibility of having a *dynamic* risk map as being highly valuable. The expectation is that AI applications will continuously collect and analyze data from past experiences, web news, communication with the customer, internal meeting minutes and informal discussions, as well as markets and financial trends to produce assessments of the expected probability of occurrence and impact of future events or conditions. An additional Knowledge Area that the experts identified as relevant to future AI applications is Project Resource Management and, particularly, allocation of project team members. It was suggested that a list of recommended team members will be created, based on sophisticated analyses of data on past experience, expertise, knowledge, skills, availability, and work relationships on the one hand, and data on the project requirements, task list, budget, and organizational priorities on the other hand.

The processes and functions selected by the panelists as being the least relevant and least important functions are presented in Table 4.

The list of less endorsed functionalities is related to project management supporting processes. Although those functionalities

Table 2. Expert Panel by Industry^a

Industry	
Software	13
Information technology	22
Professional services	15
Manufacturing	8
Construction	8
Semiconductors	3
Education	3
Healthcare	2
Other	10

^aPanelists were asked to indicate in which industry(ies) they manage(d) projects. Thus, there are more replies per category than the number of panelists.

Table 3. The Most Relevant Processes and Functions for AI in Project Management

Frequency	Knowledge Area	#	Process/Topic	Function
31	Project Schedule Management	9	Scheduling	Create project schedule
27	Project Schedule Management	13	Milestone deadlines	Analyze implications of missing deadlines
25	Project Scope Management	3	Work breakdown structure (WBS)	Create WBS/tasks list
21	Project Cost Management	14	Budgeting	Create project budget
21	Project Scope Management	8	Scope deviation	Identify scope creep and deviations
19	Project Risk Management	30	Risk identification and analysis	Produce dynamic risk map
18	Project Scope Management	1	Deliverables	Extract deliverables
14	Project Schedule Management	12	Schedule update	Update project progress and schedule
11	Project Scope Management	5	Prioritization	Prioritize tasks
9	Project Resource Management	23	Project team	Allocate team

were suggested by the panelists in previous rounds as potential AI applications for project management, when they had to prioritize them, those functions were less appreciated. Based on the experts' explanations for these selections, those functions were chosen as "least relevant" because they represent processes that should be supported by future organizational AI systems. Those processes and functions impact the daily work of project managers, but the decisions on whether and how to implement AI applications that support projects in the organizational context are typically not their responsibility. Specifically, it is worthwhile mentioning that, although the topic of selection of projects to the organizational portfolio was addressed in the previous studies on AI (e.g., Costantino et al., 2015; Liu et al., 2019; Rad & Rowzan, 2018), the above explanation can apply to this function as well if project managers do not perceive it as part of their direct role.

Interpretation and Discussion

The current exploratory study provides insights into developing the trends of applying AI solutions in businesses, especially in the context of project management. This study is based on collecting data from project management experts who, following the preliminary request to participate in the research, were enthusiastic about contributing their inputs. By using the Delphi method, an array of future potential AI applications was identified in each one of the project management Knowledge Areas. Further construct validity was employed

when the panelists were asked to validate the interpretation and categorization of their inputs (Okoli & Pawlowski, 2004). The consolidated list of 50 AI applications, categorized by project management functions in each Knowledge Area, was presented to the panelists to assign levels of relevance and importance, which resulted in a prioritization of the suggested ideas for future project management AI applications. The most preferred functionalities are related to the core processes of project management. Those processes include scope, schedule, resources, and cost management, which compose the well-known project *iron triangle*, also known as the *triple constraints* (Van Wyngaard et al., 2012), and the related risk management processes, which jointly have a major impact on project success (Martínez & Fernández-Rodríguez, 2015). Additional interesting ideas for future AI applications were suggested by the panelists in relation to each of the project management Knowledge Areas (see Appendix B for the full list) with the goals of facilitating and improving the project manager's work.

The findings reveal that the expectations of current project managers from future AI applications are mainly focused on processes typically perceived as being under their direct responsibility. Therefore, the results of this study can be used as guidelines for the development of AI applications that will be most relevant and beneficial to practicing project managers. However, although assessed as less relevant and important, additional processes were identified by the project managers as potentially enhanced by AI; those processes are usually

Table 4. The Least Relevant Processes and Functions for AI in Project Management

Frequency	Knowledge Area	#	Process/Topic	Function
15	Other	48	IP (intellectual property)	Manage innovation
14	Project Resource Management	24	Develop team	Training
14	Project Resource Management	26	Acquire resource	Recruit employees
8	Project Resource Management	25	Manage team	Assess team performance
8	Other	47	Security	Fraud inspection
7	Project Communications Management	27	Plan communication	Plan meetings
7	Project Stakeholder Management	37	Contact list	Create contact list
6	Project Resource Management	22	Knowledge management	Create knowledge maps
6	Other	50	Portfolio	Select projects

aimed at supporting the project manager's work and performed on the organizational level. Among those processes, one can find recruitment and training of employees, selecting vendors, negotiating with subcontractors, management of intellectual property, and selection of projects for the organizational portfolio. These findings imply that many project managers nowadays have a wider perspective. They are expected to go beyond their traditional responsibilities, to know more, and be more involved in the organization's strategy and achievement of its goals, as demonstrated in the PMI Talent Triangle® (PMI, 2015) and in the International Project Management Association's *Individual Competence Baseline 4.0* (International Project Management Association [IPMA], 2015). Both standards refer not only to the technical or practice competencies but also to people and leadership, as well as to the perspectives of business management and strategy. This is in line with the conceptual framework that describes the current and future project manager as "chief executive of the temporary organization ... and the agent of the principal" (Turner & Müller, 2003, p. 2).

Considering the expected high volume and impact that AI will have on organizations (Iansiti & Lakhani, 2020; Raisch & Krakowski, 2020) and projects (Auth et al., 2019; Wang, 2019), there are some thought-provoking issues that should be addressed prior to developing and implementing AI applications in projects.

The Future Roles of Project Managers

The current study identifies different functionalities that can be beneficial for project managers, especially those perceived as the basic processes in planning and managing projects and involve judgment and decision-making. A previous study mentioned administrative work as the first level for AI systems to save time and money, whereas further stages identify practices such as design thinking and social skills that successful future managers will need to master (Kolbjørnsrud et al., 2016). The results of the current Delphi study indicate that project managers recognize the primary need to address the basic processes, but they are also open to accepting, in further stages, enhanced technologies that will improve their work as leaders. This is in line with findings of a study by Anantatmula who claims that people-related issues and leadership are core factors in project management, and that technology can help project managers promote performance, team development, and competency (Anantatmula, 2008). In a seminal work by Robert Katz, an administrator was defined as "one who (a) directs the activities of other persons and (b) undertakes the responsibility for achieving certain objectives through these efforts" (Katz, 1974, p. 6; 2009). Katz distinguished among three basic interrelated skills: technical skills, human skills, and conceptual skills. For project managers, technical skills relate to understanding and proficiency with regard to a specific activity, based on relevant methods, processes, procedures, or techniques. Human skills relate to the abilities to effectively work in a team and lead cooperative efforts, whereas conceptual skills relate to envisioning the project as a whole and understanding

the relationships among different factors in the project environment and beyond (El-Sabaa, 2001). Looking into the future role of project managers it seems that the activities based on technical skills can be replaced by AI applications, whereas the activities based on human and conceptual skills will be enhanced by AI applications. Furthermore, thanks to AI applications, project managers will be able to spend more time and effort in managing the interpersonal and strategic activities. They will be more focused on applying soft skills such as communication and conflict resolution, whereas the AI systems will help with data-associated tasks and the more technical elements.

Taking into consideration the central role of decision-making in the daily work lives of project managers, the results indicate that AI is expected to support and improve those processes. Recent discussions on AI and its impact on managers and managerial practices have led to the understanding that AI is and will be complementary to human activities in decision-making situations, especially in uncertain, complex, and equivocal environments (Jarrahi, 2018), but will not replace humans (Barro & Davenport, 2019; De Cremer & David Kasparov, 2021; Wilson & Daugherty, 2018).

A review of the suggested AI applications in this study shows that, in addition to information that will be analytically processed using the data collected from multiple sources on a specific topic, 40 items (out of 50) also indicate that there is an expectation for AI applications to make recommendations to project managers. This means that the results of analyzed data will include inferences based on a deduction process that follows a set of predefined and developing rules, while considering relevant circumstances. Moreover, 17 items on the list of potential AI applications for project management include actionable components, which refer to actual activities that will be carried out independently by the applications. Therefore, even if it takes some time to achieve mature AI systems that work properly in a project environment, it can be concluded that the role of project managers is expected to change dramatically. Many of the tasks that project managers are doing these days will be efficiently completed by computers that will apply cognitive intelligence, whereas project managers will be required to demonstrate advanced types of competencies that are more related to emotional intelligence and social intelligence (Kaplan & Haenlein, 2019).

Ethical Issues

Although the current study was focused on professional processes and functions in the ten project management Knowledge Areas, an additional issue—ethical implications—has spontaneously emerged from the participants' responses. The development of AI technologies may lead, not only to augmentation of human tasks, and particularly project managers' activities in the workplace, it also suggests the replacement of employees and managers, which raises major concerns. A study among 550 AI experts on the future of high-level

machine intelligence and superintelligence (intelligent systems that exceed humans' cognitive performance) concluded that "superintelligence is likely to come in a few decades and quite possibly bad for humanity" (Müller & Bostrom, 2016, p. 15). Similar concerns were raised by some of the current study expert panelists, who expressed a fear that there will be systems capable of completing these tasks quicker and more accurately than humans in the near future. The debate on the feasibility of superintelligence, in other words, machine agents that surpass and exceed those of the human mind—both on the conceptual and computational levels—has drawn a lot of attention over the last decades. Although there is no consensus on if and when it will be possible to develop superintelligence systems, many scholars argue that it is inherently impossible (e.g., Alfonseca et al., 2021; Jebari & Lundborg, 2020), and there is broad agreement that the current state of the art in AI cannot support superintelligence. Most of the panelists found the collective list very exciting and inspiring. Alongside the positive benefits that AI may offer to projects, however, the participants expressed their worries about the negative impact that might come with AI in terms of social interactions, ethics, and privacy, which were also mentioned in other studies on AI (Bailey & Barley, 2020; Boström & Yudkowsky, 2014; Dwivedi et al., 2019). Specifically, the panelists expressed their concerns with regard to the availability of data, including personal information and analyses of behaviors that might impact project-related decisions in the domains of team management, negotiation, and communications.

Conclusions

This study explored the future of AI in projects from the perspectives of project managers. It is a pioneering examination of how AI is expected to reshape project management in the future by offering solutions that will improve and enhance the tasks that are currently carried out by project managers and their teams. The findings from this study have theoretical and practical implications. For scholars, this is the first study that comprehensively identifies possible future AI applications based on the Delphi method following a rigorous three-round group decision-making approach. The results of the study categorize relevant items in each of the project management Knowledge Areas, thus providing a useful and organized basis for future research on AI in each one of these areas. In addition, the most important functions to be supported by AI were identified in this study and can be used as outlines for further studies that will benefit the project management profession. The top-ranked functions are: create a project schedule, analyze implications of missing deadlines, create a WBS/tasks list, create a project budget, update project progress and schedule, identify scope creep and deviations, produce a dynamic risk map, extract deliverables, prioritize tasks, and allocate team members.

From a practical perspective, the results of the study can be exploited by organizations that manage projects to decide in which AI systems and applications to invest to gain a competitive advantage. Although not all the technologies needed to

successfully complete those tasks are currently available, according to a survey by Gartner, among 1,070 CIOs, 52% have already deployed or plan to deploy emerging technologies such as AI in the next 12 months (Gartner, 2020). The insights provided by this study can be used by organizations to direct their resources toward AI solutions that are the most relevant and beneficial for their project managers.

This study is not free of limitations. First, the Delphi method has several weaknesses and, therefore, been criticized by researchers (Landeta, 2006; Woudenberg, 1991) and not extensively used in management studies (Grisham, 2009). However, this study is based on a large panel of experts, which is wider than recommended in the literature (Hasson et al., 2000; Sekayi & Kennedy, 2017). The large number of participants and their levels of expertise reinforce the study's validity. Second, the study has focused on exploring relevant AI applications in project management based on the opinions of project managers, without input from AI experts. The expert panel is based solely on the prospective users' expectations and does not represent the perspectives of practitioners in AI technologies, which has been examined in other studies (e.g., Darko et al., 2020; Martínez & Fernández-Rodríguez, 2015). The decision to include in the expert panel only project management experts derived from the intention to allow contributions free of limitation, which does not consider the availability of technologies, but rather gets inspiring ideas for future developments that respond to the users' requirements and needs.

Future studies can build on the foundations presented in this article to examine appropriate ways and corresponding technologies to develop specific AI solutions for each of the project management Knowledge Areas or a combination of several Knowledge Areas. Another venue for further research would be to examine what other stakeholders in project organizations expect from AI, and how AI experts perceive potential future applications for the benefit of better project management. In addition, this study can be considered a starting point to investigating if there are different needs for AI applications for projects in different industries.

Appendix A: An Introductory Letter to the Participants

Dear XXXX,

Thank you for accepting our invitation to take part in a three-round survey to explore the future of artificial intelligence (AI) in project management.

Your participation will be completely confidential, and your responses will remain completely anonymous throughout this process; however, if you wish to be publicly accredited as a participant in the study, please send us a confirmation by email stating that once the research results will be published (in journals, conferences, etc.) you agree to be identified as one of the participants.

The data gathered within this survey will not be subject to any public disclosure and are for use only as part of a research project.

The overall research aims to propose a framework for AI in the context of project management.

AI is defined as “systems that display intelligent behavior by analyzing their environment and taking actions—with some degree of autonomy—to achieve specific goals” (European Commission, “Artificial Intelligence for Europe”). Specifically, the current study wishes to identify potential outputs of AI applications on three levels: information, recommendation, and actions.

Information refers to the presentation of collected data from multiple sources on a specific topic.

Recommendation refers to results of analyzed data where inferences are made based on a deduction process that follows a set of predefined and developing rules and considers relevant conditions.

Actions refer to actual activities that will be carried out independently by the applications.

The following survey is the first stage of a Delphi questionnaire. It is designed to obtain your personal opinion regarding the research topic.

The Delphi process involves three rounds of questioning on three separate occasions with 20 to 30 days between each round.

Round 1: General open-ended questions where you are requested to provide your response. These questions are presented below.

Round 2: The answers that will be collected from you and all the other panelists (project management experts), will be consolidated and summarized into a set of specific questions in the form of statements that you will be asked to review, comment on, and further elaborate and build upon with new statements.

Round 3: The responses that will be collected from you and all the other panelists will be analyzed and presented to you again in an orderly manner. You will be able to see the overall and average responses and you will be asked about your level of agreement.

The identity of all panelists will remain confidential at all times, unless otherwise approved by you in writing.

We appreciate your contribution and remain at your disposal for any additional information.

Thank you,

The Researchers

Appendix B: AI Applications for Project Management

#	Process/Topic	Function	Details	Data Sources
Project Scope Management				
1	Deliverables	Extract deliverables	Identify deliverables, contractual dates, constraints, expectations, costs, quality standards, reporting procedures.	SOW; charter; meetings; previous projects both within the organization and external ones; similar products/ services; standards; templates; organizational procedures; financial records; web data; observations (pictures); maps; RFI/RFP
2	Expected deliverables	Verify deliverables	Identify existing similar components of deliverables (in the organization or outside).	previous projects both within the organization and external ones; similar products/services; synthetic data; SOW; contract
3	WBS	Create WBS/tasks list	Identify all needed tasks, create work plan, composed of high-level and low-level working units, based on deliverables, and estimate efforts and costs and recommend on assignment of resources.	SOW; RFI/RFP; technical data; HR data; financial data; previous projects; web data; synthetic data; documented retrospectives
4	Product backlog	Create user stories	Follow users' everyday work, score the processes frequency/duration of each step/user needs, and suggest a list of user stories.	observations; meetings; discussions; mail communication; video conferences discussions, similar projects' data; internal project documentation
5	Prioritization	Prioritize tasks	Create dynamic prioritized backlog, and help prioritize project tasks.	meetings with customers; market trends; contracted vendors; resource availability; previous projects
6	Requirements	Analyze requirements	Identify all requirements and analyze by level of importance, including recommendations on elimination of unnecessary requirements. Gather all cost	SOW; similar projects, emails, meeting minutes, web data, internal and external communication and information

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#	Process/Topic	Function	Details	Data Sources
7	Specifications	Update requirements	and schedule data for each one and the expected ROI of adding it. Produce requirements traceability report, including recommendations on managing gaps, update change requests.	systems; innovations in similar products worldwide; regulations and standards product specifications; team meetings; one note; ticketing systems; emails; conference calls;
8	Scope deviation	Identify scope creep and deviations	Identify and alert on scope creep and deviations from plan, and analyze implications.	email exchange; meeting minutes; risk register; status reviews; burndown charts, budget reviews
Project Schedule Management				
9	Scheduling	Create project schedule	Smart schedule planning, based on feature complexity, available resources, team expertise, constraints, etc. (including duration and number of sprints/task duration and dependencies, etc.);	past experience; previous projects; specifications, HR data; team calendars; international and local events; best practice knowledge base
10	Buffers	Set buffers	analyze work plan, previous projects and external data to recommend and set buffers.	past experience; project plan; external data; project risks
11	Milestones	Set project milestones	Identify patterns to recommend and set project milestones and gates.	SOW; work plan, previous projects data, communication with customer, contract, deliverables; company standards
12	Schedule update	Update project progress and schedule	Update/suggest updates to project schedule; identify current and expected deviations from schedule, analyze implications, and notify relevant stakeholders.	project reports; external events; web data; resource availability; emails, conversation during status meetings, and any informal communications among project stakeholders
13	Milestones deadlines	Analyze implications of missing deadlines	Identify expected deadlines that will not be met, analyze the impact (future schedule, penalties, benefit deferrals, cost impact, etc.), inform the project manager, and recommend actions to avoid this.	SOW; agreement; meetings; external dependencies; correspondences; standards; resource availability, inner organizational knowledge, project expected benefits, cash flow implications
Project Cost Management				
14	Budgeting	Create project budget	Create initial budget suggestion, analyze collected data on cost drivers, analyze financial trends, and identify potential savings.	past experience; financial records; HR data; similar projects; web data (tax rates, exchange rates, etc.); market prices; synthetic data; procurement/organizational databases;
15	Budget plan	Manage reserves	Analyze project risk data to recommend management of reserves and emergency reserves.	work plan, risk plan; organizational procedures; previous projects; data on customer; web data
16	Budget update	Update project budget	Identify changes in cost drivers and update the project budget; identify expected cost overrun, identify opportunities for savings, analyze implications, produce forecasts, and recommend actions.	cost changes in markets; official financial sites; email exchange; meeting minutes; risk register; status reviews; web data
Project Quality Management				
17	Quality plan	Create quality plan	Analyze deliverables to recommend an optimal quality assurance plan based on industry standards, cost/schedule targets, related risks.	work plan, specifications, standards, on-site supervision; project reports
18	KPI	Set KPI for the project	Set/suggest project KPI.	previous project KPI; industry standards; expected benefits; project constraints
19	Quality control	Generate test scripts	Analyze business requirements to produce test scripts and update the scripts based on input collected throughout the project.	project documents; business strategy, meeting minutes, user stories; standards
20		Assess product quality	Run automatic validation tests of the system	SOW; specifications; requirements;

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#	Process/Topic	Function	Details	Data Sources
	Quality assessment		and produce report; analyze product quality; identify upper limit/lower limit, and recommend corrective actions.	standards; use cases; live data collection by cameras, drones; 3D scanning products; reported issues log; monitoring tools (reporting issues, crashes, defects, etc.)
21	Quality assurance	Assess project effectiveness	Help analyze project processes and recommend corrective actions, or automatically generate questionnaires to do so.	communication with customer; product requirements; expectations
Project Resource Management				
22	Knowledge management	Create knowledge maps	Identify stakeholders' explicit and implicit knowledge on topics and produce a knowledge map. Advise stakeholders on who they can turn to with different topics.	HR data; social networks, emails, meeting minutes; informal meetings; information from outside the organization
23	Project team	Allocate team	Create a list of recommended team members for a project based on project needs, required skill set, personalities, interpersonal interactions, availability, organizational priorities, preferences of team members.	past experience; informal communication; project plan; organizational resource pool, resource plan, training databases, skills in LinkedIn, personality tests, simulations, defined R&R
24	Develop team	Training	Analyze employees' competences and needs to recommend on personal/team/organization trainings.	employee CVs; employee request; periodical assessment; report on activities; industry standards; organizational procedures; employee achievements in previous projects, duration of tasks, and analysis of communication skills of the employees
25	Manage team	Assess team performance	Analyze employees' behavior and performance to provide feedback and recommendations on career development.	observations; formal and informal communications; status review; time to complete tasks, time it takes to reply to emails, peer reviews; information about employees' personal status
26	Acquire resource	Recruit employees	Proactive search for filling open positions and finding experts based on CV collection and screening and BOT (an autonomous digital program that can interact with users), interview (automatic headhunter).	social media; organizational resource pool; HR data; grades of professional and academic courses; online testing for character matches and skill sets
Project Communications Management				
27	Plan communication	Plan meetings	Set the meeting agenda based on work plan and current status and invite relevant stakeholders.	SOW; work plan; project reports; observations; emails; other meetings
28	Manage communication	Summarize meetings	Produce meeting minutes, including identification of decisions and actions items, and follow up on action items.	formal meetings; recording; organizational structure; work plan; schedule
29	Manage communication	Inform stakeholders	Analyze communications to identify patterns and detect relevant stakeholders who should be notified on issues; send notifications and reports and messages to relevant stakeholders with relevant project information and/or action items where needed; identify how various stakeholders prefer communications. Categorize emails by subjects/folders to help stakeholders focus.	emails; reports; meetings; formal and informal communications; work plan

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#	Process/Topic	Function	Details	Data Sources
Project Risk Management				
30	Risk identification and analysis	Produce dynamic risk map	Collect data on internal and external risks and update the risk plan, including assessments of probability and impact.	past experience, web news; customer expectations; team skill set; project status; project environment analysis
31	Risk response	Produce dynamic risk mitigation plan	Analyze and evaluate data on identified and emerging risks to recommend actions (avoidance, acceptance, etc.) and relevant stakeholders who should weigh in.	SOW; project plan; formal and informal communications; web data; news; HR data; market trends; financial trends
Project Procurement Management				
32	Plan procurement	Select potential vendors/subcontractors	Collect data on vendors, suggest potential vendors, compare offers regarding costs, quality, availability, reputation, etc.	past experience; RFI/RFPs and bids; general information on the internet; SOW; work plan; schedule; budget; vendor information systems
33	Negotiations	Negotiate with vendors/subcontractors	Collect data on selected vendors and competitors to identify weaknesses and strengths to recommend strategies of negotiation.	web data; previous projects; financial records
34	Contracts	Prepare contract	Generate recommended contract outline based on standards and vendor negotiations.	SOW; regulations; standards; legal department guidelines; emails; documents; meetings
35	Conduct procurement	Analyze timing for procurement	Analyze internal and external data to recommend the optimal timing for purchases based on project data, market trends, and exchanges.	web data; work plan; progress reports; financial records; other projects in the organization; stakeholder correspondence
Project Stakeholder Management				
36	Identify stakeholders	Create stakeholder map	Identify relevant and perceived, internal and external stakeholders in the project environment, their required and expected involvement, and their impact on project success.	past experience; organizational procedures; formal and informal communications; web data
37	Contact list	Create contact list	Create and update contact list, including changes in team personnel, customers, vendors, and contractors, automatically changing meeting invitations and mailing lists to include the right lists and update them accordingly.	correspondences; meetings; social media; HR records
38	Manage engagement	Manage stakeholders	Analyze stakeholder communication and assess engagement and morale to recommend actions to manage stakeholders (keep them satisfied, involved, monitored, informed, and managed closely).	formal and informal communications; stakeholder body language, responsiveness to emails, reports, meeting invitations, update requests
39	Manage engagement	Improve collaboration	Analyze stakeholders' interests, activities and experience to identify potential opportunities for collaboration.	HR database; social media; work plan; stakeholder surveys
Project Integration Management				
40	System complexity	Identify and analyze interfaces	Analyze project components, detect other systems that may impact or be impacted, and notify the project manager and relevant stakeholders.	organizational data; formal and informal communications; previous projects; similar products; web data; project design plan (architecture, specification documents)
41	Change management	Change requests	Analyze communication data to identify change requests and recommend which stakeholders need to be involved and finally recommend on a decision (approve/defer/reject); analyze potential impact the change may have on the project.	internal communications; correspondence with the customer

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#	Process/Topic	Function	Details	Data Sources
42	Progress control	Analyze and update progress and produce earned value assessments	Analyze integrated project data, update project plan, produce earned value analysis, estimate completion, and recommend corrective actions.	SOW; team meetings; standards; visual analysis of the product
43	Business value	Analyze business value	Analyze planned and actual team efforts, resource utilization and costs in comparison with project business objectives to identify deviations (gold plating) and recommend actions.	organization strategy; SOW; charter; project reports
Other				
44	Project success	Assess project success	Create KPIs by integrating internal information and data on the customer and market and assess project effectiveness and efficiency. Provide forecasts on project success.	SOW; contract; project reports; formal and informal communications; historic measurements, potential risks; market trends; competitor offerings
45	Lessons learned	Analyze lessons learned	Produce lessons learned reports that integrate data from current project and similar projects, and send them at the right time to various stakeholders.	project information (quality and budget information, plan vs. actual performance); organizational archive; retrospectives; team meetings; incidents investigations
46	Safety	Identify safety issues	Identify issues of safety management and inform relevant stakeholders.	organizational procedures; standards; legal documents; work plan; responsibility matrix; observations; live data collection by using cameras, drones, satellites
47	Security	Fraud inspection	Identify exceptional actions/transactions and inform relevant stakeholders.	financial records; budget; organizational procedures; legal documents; work plan; responsibility matrix
48	IP (intellectual property)	Manage innovation	Screen requirements and specifications and search for patents related to the product; inform relevant stakeholders.	requirements; specifications; web publications on patent applications and registrations
49	Make decisions	Crisis management	Simulate one or all the six hats (Eduard de-Bono) and/or “the devil’s advocate” to generate recommendations to the project manager.	SOW; work plan; schedule; budget; HR data; risk register
50	Portfolio	Select projects	Analyze organizational strategy, market, and technology to recommend project selection.	organizational records; web data on market trends and competitors; social media

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